

Short Channel CNT paper replication

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Abstract

In this tutorial, we derive analytically the schottky barrier height for a CNT tube that matches rigorous non-equilibrium Green's function simulations [1].

1 Introduction

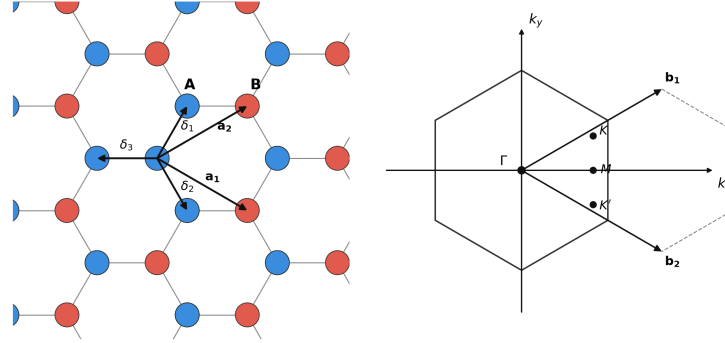


Figure 1: Graphene lattice and its Brillouin zone. [2]

Given the above coordinate choice for graphene unit cell, the band structure of a (N,0) CNT, obtained from the tight-binding model of graphene [2], is expressed as:

$$E = \pm t_0 \sqrt{f(k_x, m)} = \pm t_0 \sqrt{1 + 4 \cos\left(\frac{3k_x a}{2}\right) \cos\left(\frac{\pi m}{N}\right) + 4 \cos^2\left(\frac{\pi m}{N}\right)} \quad (1)$$

where $t_0 = 2.5\text{eV}$ is the tight-binding coupling constant and m is an index that runs from 0 to $N - 1$.

1.1 Analytical expression for band-gap

CNT is one of those nice systems where the band-gap can be expressed analytically. In (1), consider the bands with a + sign. Then the band-minimum, E_{min_m} for a given band index, m , occurs when

$$\partial_{k_x} f(k_x, m) = 0 \implies \frac{3k_x a}{2} \in \{0, \pi\} \quad (2)$$

while the minimum of the set E_{min_m} across all bands, as m is varied is obtained when the quadratic equation in $\cos(\frac{\pi m}{N})$ is minimized:

$$\begin{aligned} \partial_{\cos(\pi m/N)} f(k_x, m) \Big|_{\frac{3k_x a}{2} \in \{0, \pi\}} &= 0 \\ \implies \cos\left(\frac{\pi m}{N}\right) &= -\frac{1}{2} \cos\left(\frac{3k_x a}{2}\right) \Big|_{\frac{3k_x a}{2} \in \{0, \pi\}} \\ &= \pm \frac{1}{2} \end{aligned} \quad (3)$$

which means $\cos\left(\frac{\pi m}{N}\right)$ needs to be as close to $\pm \frac{1}{2}$ as possible. This is exactly possible whenever $N = 3m$, then band-gap is exactly 0 and we call such a CNT metallic. Though in this article, we will consider only CNTs with a bandgap so let us restrict ourselves to when $N = 3m - 1$, i.e $N = 8, 11, 14, 17$ etc.

Given a value of N , m will take values in the set $\{0, 1, 2, \dots, N-1\}$. Let

$$m_0 := \arg \min_m \left| m - \frac{N}{3} \right|.$$

The band-minima is then written as (pairing the value of m_0 with $\frac{3k_x a}{2} = \pi$):

$$E_g = 2t_0 \sqrt{1 - 4 \cos\left(\frac{\pi m_0}{N}\right) + 4 \cos^2\left(\frac{\pi m_0}{N}\right)}. \quad (4)$$

1.2 Analytical expression for Density of States

Good point to obtain an expression for the density of states near the band-minimum. First, we need to show the following approximation holds near the band-minimum:

$$\frac{d}{dE} \left(\frac{3k_x a}{2} \right) \approx \frac{|E|}{t_0 \sqrt{E^2 - (E_g/2)^2}} \quad (5)$$

In section 1.1 we discovered that a neighbourhood of band minimum comprises of m_0 with $\frac{3k_x a}{2} \approx \pi$. So equation (1) around band-minimum can be

rewritten as a taylor expansion around $\frac{3k_x a}{2} = \pi$: (Hereafter, k_x refers to the small-signal expansion around $k_x = \frac{2\pi}{3a}$)

$$E \approx \pm t_0 \sqrt{1 + 4 \cos\left(\frac{\pi m_0}{N}\right) \left(-1 - \frac{1}{2} \left(\frac{3k_x a}{2}\right)^2\right) + 4 \cos^2\left(\frac{\pi m_0}{N}\right)} \quad (6)$$

Plugging in the value of E_g from (4), we get:

$$E \approx \pm t_0 \sqrt{\left(\frac{E_g}{2t_0}\right)^2 - \left(2 \cos\left(\frac{\pi m_0}{N}\right)\right) \left(\frac{3k_x a}{2}\right)^2} \quad (7)$$

Since m_0 takes $\cos(\frac{\pi m_0}{N})$ to be as close to $\frac{1}{2}$ as possible, for a big enough N , we can approximate the above equation as:

$$E \approx \pm t_0 \sqrt{\left(\frac{E_g}{2t_0}\right)^2 - \left(\frac{3k_x a}{2}\right)^2} \quad (8)$$

Rerrangment gives:

$$\frac{3k_x a}{2} \approx \pm \frac{1}{t_0} \sqrt{E^2 - \left(\frac{E_g}{2}\right)^2} \quad (9)$$

(5) follows from differentiating (9) with respect to E .

References

- [1] Francois Leonard and Derek A. Stewart. Properties of short channel ballistic carbon nanotube transistors with ohmic contacts. *Nanotechnology*, 17(18):4699–4705, 2006. arXiv:cond-mat/0608616.
- [2] A. H. Castro Neto, F. Guinea, N. M. R. Peres, K. S. Novoselov, and A. K. Geim. The electronic properties of graphene. *Reviews of Modern Physics*, 81(1):109–162, 2009. arXiv:0709.1163.